

# Finite order obstructions for edge-resilient four-critical graphs

Rachel Teo<sup>1</sup>, Wenyi Wang<sup>1</sup>, Hamdi Abderrahmene<sup>1</sup>, Alejandro Zarzuelo Urdiales<sup>2</sup>

**Affiliations:** <sup>1</sup> Sakana AI; <sup>2</sup> Open Math.

[Code and formal proofs](#)

## Abstract

A finite vertex-critical four-chromatic graph that stays four-chromatic after deletion of any  $r$  or fewer edges satisfies  $27r+19 \leq 4n$ , together with stronger quadratic and integral independent-set filters. These refine explicit finite lower bounds, without improving the prior eventual asymptotic restriction.

## Graph hypotheses and inequalities

Let  $G$  be a finite simple graph on  $n$  vertices, with chromatic number four. Assume that deletion of any vertex lowers that number, and that deletion of any set of at most  $r$  edges does not lower it, where  $r \geq 1$ . The submitted theorem gives  $27r+19 \leq 4n$ .

Two further necessary conditions are  $27(r+1)(n-1) \leq (2n+1)^2$  and  $9b(r+1) \leq (2b+1)(n-b)$ , where  $b = \lfloor (n+1)/3 \rfloor$ . The integral filter can exclude an order allowed by the linear inequality: at  $r=4$  it requires  $n \geq 33$ , whereas the linear inequality alone gives  $n \geq 32$ . These are one family of obstructions, not three separate original results.

## Criticality supplies a finite code

Deleting a vertex of a four-critical graph produces a three-colouring of the remainder. The proof extracts a large independent colour class and encodes neighbourhood information on that class. Edge resilience forces separation between the relevant codewords: too few differences would produce a critical edge set of size at most  $r$ .

The finite-code theorem compares a lower bound for total pairwise distance with an upper bound obtained by coordinate counts. This is a Plotkin/Cauchy-Schwarz argument. Optimizing the independent-set parameter and retaining its integer size gives the quadratic and exact filters; a final numerical derivation yields the clean linear inequality.

The formal graph module supplies every hypothesis of the numeric code lemma from the actual canonical criticality definitions. The final theorem is not a numeric inequality assuming an unproved graph-to-code model. The target audit checks deletion and finite-cardinality conventions against the registered problem.

## An ordinary derivation of the code and order bounds

Put  $s=r+1 \geq 2$ . For any vertex  $v$ , choose a proper three-colouring of  $G-v$ . Each of its three colour classes contains at least  $s$  neighbours of  $v$ : otherwise delete the fewer than  $s$  edges from  $v$  into that class and give  $v$  its colour, producing a three-colouring after too few deletions. Thus every vertex has degree at least  $3s$ .

Choose a maximum independent set  $A$  of size  $a$  and put  $H = V(G) \setminus A$ ,  $|H| = h = n - a$ . A three-colouring after deleting one vertex shows  $n - 1 \leq 3a$ . For each  $v \in A$  let  $C_v \subseteq H$  be its nonneighbours in  $H$ . Independence of  $A$  and the degree bound imply  $|C_v| \leq h - 3s$ . If  $u, v \in A$  are distinct, take a three-colouring of  $G - u$  and look at neighbours  $x$  of  $u$  having the colour of  $v$ . There are at least  $s$  of them. Such an  $x$  is outside  $A$  and cannot be adjacent to  $v$ , so it belongs to  $C_v \setminus C_u$ . Therefore  $|C_v \setminus C_u| \geq s$  in both directed orders.

Here is the complete numerical lemma. Let a subsets  $C_v$  of an  $h$ -element ground set satisfy those weight and directed separation bounds. Write  $t_x = \#\{v : x \in C_v\}$ ,  $T = \sum t_x$  and  $Q = \sum t_x^2$ . Double counting directed differences gives  $a(a-1)s \leq \sum_{(u,v)} |C_v \setminus C_u| = aT - Q$ . Also  $T \leq a(h-3s)$  and  $T^2 \leq hQ$  by Cauchy-Schwarz.

If  $h \leq 6s$ , then  $T \leq a(h-3s) \leq ah/2$ . The function  $aT - T^2/h$  is increasing for  $0 \leq T \leq ah/2$ , so  $a(a-1)s \leq a^2(h-3s) - a^2(h-3s)^2/h = 3a^2s(h-3s)/h$ . Rearrangement gives  $9as \leq (2a+1)h$ . If  $h > 6s$ , that inequality is immediate from  $(2a+1)h > 6s(2a+1) > 9as$ . This proves the code bound in all cases. The separation of two codewords also gives  $h \geq 4s$ , because one  $C_v$  has at least  $s$  members while its weight is at most  $h - 3s$ .

The degree bound gives  $n \geq 3s + 1 \geq 7$ ; hence  $a \geq \lceil (n-1)/3 \rceil \geq 2$ , so two codewords do exist. Combining  $h \geq 4s$  with  $n - 1 \leq 3a$  and  $n = a + h$  yields  $2n \geq 12s - 1$  and therefore  $n \geq 6s$ . Now divide the code bound by  $a > 0$ . The function  $(2a+1)(n-a)/a = 2n - 2a + n/a - 1$  is decreasing in positive  $a$ . Substituting the smaller integer  $b = \lfloor (n+1)/3 \rfloor$  gives  $9bs \leq (2b+1)(n-b)$ . Substituting the real lower value  $(n-1)/3$  instead gives  $27s(n-1) \leq (2n+1)^2$ .

Finally  $n \geq 6s \geq 12$ . If  $27s \geq 4n + 9$ , the quadratic inequality would force  $(4n+9)(n-1) \leq (2n+1)^2$ , or  $n \leq 10$ , a contradiction. Therefore  $27s \leq 4n + 8$ . Replacing  $s$  by  $r+1$  yields  $27r+19 \leq 4n$ , as claimed. This derives all three displayed inequalities from the graph hypotheses, without an assumed coding model.

## What the finite improvement accomplishes

The located prior explicit bound is  $n \geq 5r + 6$ . After integer rounding, the new linear condition ties at  $r=1$  and improves the finite lower requirement for larger  $r$ . However, Skottova and Steiner also prove an eventual restriction  $r = O(n/(\log n)^c)$ , which is stronger asymptotically. The new theorem does not supersede that result or construct any graph meeting its obstruction.

In particular it does not solve existence for  $r \geq 2$ , cover  $k > 4$ , or address infinite graphs. Known  $r=1$  constructions are prior evidence, not a new contribution of this packet.

## Formal scope and prior comparison

The three proof modules and canonical target audit were checked from the frozen sources using Lean and standard kernel axioms. Their exact compiler/library pins and declarations are supplied in the companion repository.

The finite inequalities follow from criticality-to-code separation and classical finite coding estimates. They refine explicit finite order bounds while preserving the distinction from the stronger eventual asymptotic restriction of Skottova and Steiner.

## Formal verification

The selected canonical linear, quadratic and integral order bounds have a completed Lean replay of the three-source closure. This verifies the stated necessary conditions, not existence of resilient critical graphs. Exact proof files and audit records are supplied in the companion repository.

## Erdos944Bounds.erdos944\_order\_linear

```
theorem erdos944_order_linear {r : ℕ} (hr : 1 ≤ r)
(hG : G.IsCritical 4)
(hE : ∀ edges : Set (Sym2 V), G.IsCriticalEdges edges → r < edges.ncard) :
27*r + 19 ≤ 4 * Fintype.card V
```

Source: proofs/erdos944-order-bound/OpHack/Erdos944Bounds.lean:42.

Source SHA-256: 8f9e544d0e9f465f8d92d0c0b878ab2bd9b542c0c969eaa4e2d7c908accec27f4

## Erdos944Bounds.erdos944\_order\_quadratic

```
theorem erdos944_order_quadratic {r : ℕ} (hr : 1 ≤ r)
(hG : G.IsCritical 4)
(hE : ∀ edges : Set (Sym2 V), G.IsCriticalEdges edges → r < edges.ncard) :
27 * (r+1) * (Fintype.card V - 1) ≤ (2 * Fintype.card V + 1)^2
```

Source: proofs/erdos944-order-bound/OpHack/Erdos944Bounds.lean:28.

Source SHA-256: 8f9e544d0e9f465f8d92d0c0b878ab2bd9b542c0c969eaa4e2d7c908accec27f4

**Sources and attribution:** 1. [www.erdosproblems.com](http://www.erdosproblems.com/) / 944; 2. arXiv source: 2508.08703; 3. [google-deepmind/formal-conjectures](https://github.com/google-deepmind/formal-conjectures) / 944.lean; 4. [KitaKen1/erdos-944-dirac-k4-lean](https://github.com/KitaKen1/erdos-944-dirac-k4-lean); 5. arXiv source: 2606.18462; 6. [alejandroz/ulam-opdp-difficulty-atlas](https://github.com/alejandroz/ulam-opdp-difficulty-atlas); 7. [alejandroz/openmath-2026-judging](https://github.com/alejandroz/openmath-2026-judging) / E02-erdos944-order-bound.txt; Proof source repository.

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